

Printer Dover

Spruce version 11.2 -- spooler version 11.2

File: MemC-apcRev-Be.sil etc.

Creation date: 31-Jul-85 11:29:41

For: Spruce

26 total sheets = 25 pages, 1 copy.

Problems encountered:

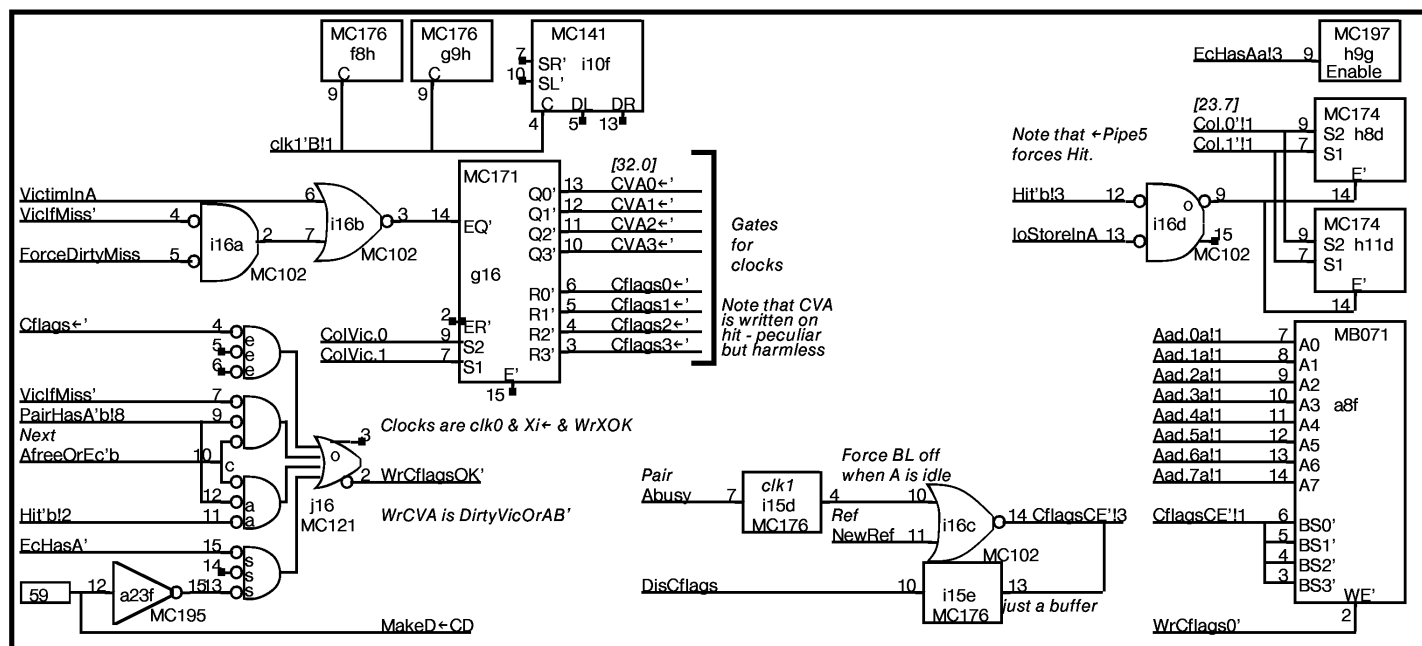
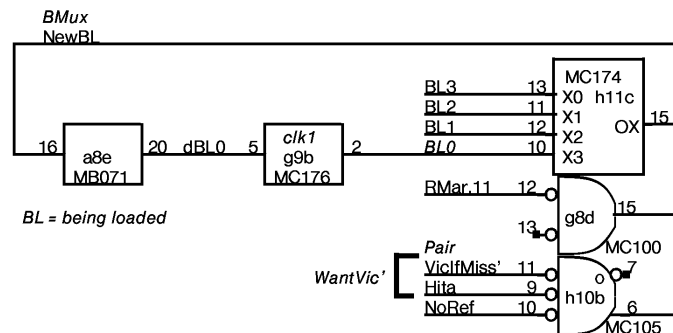
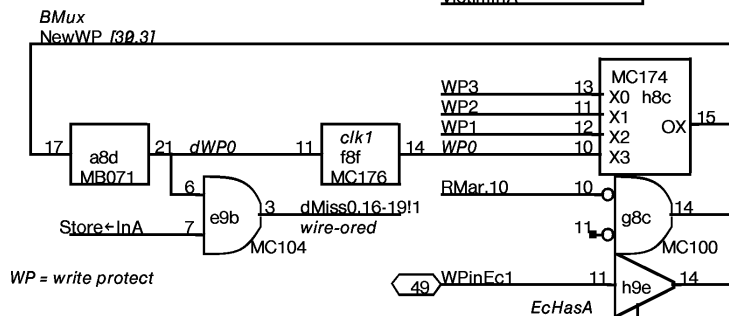
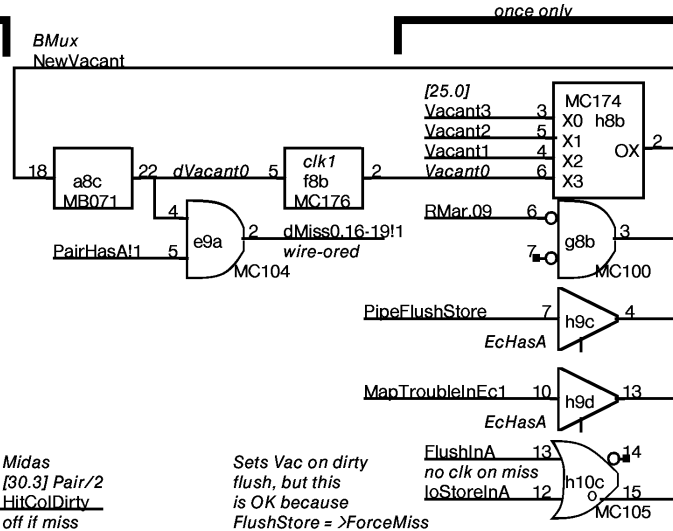
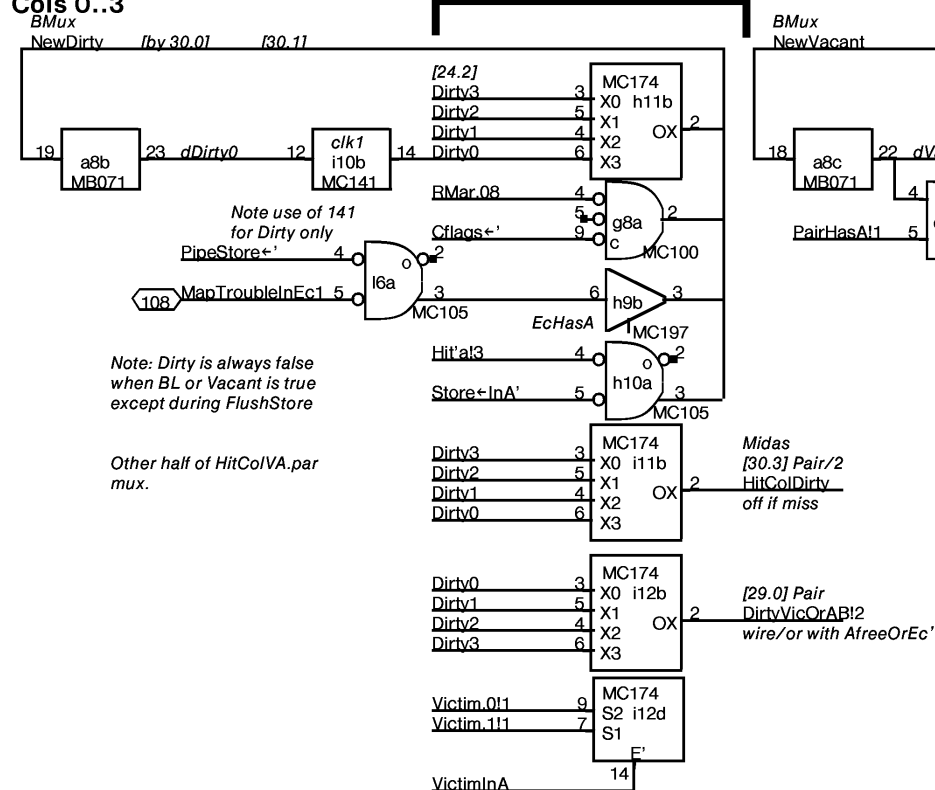
Font HELVETICA24B substituted for font HELVE24.

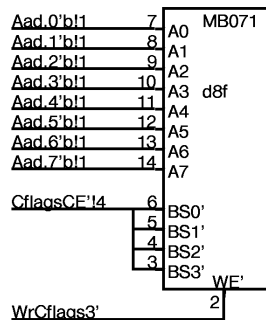
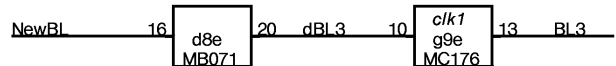
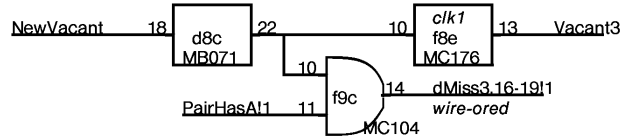
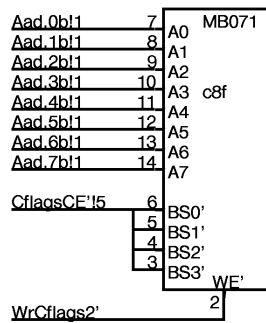
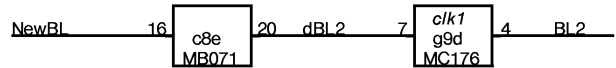
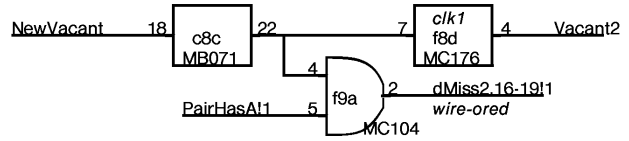
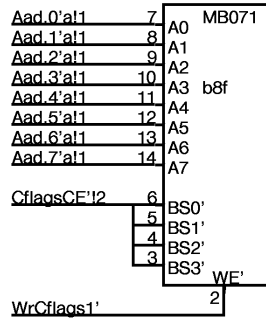
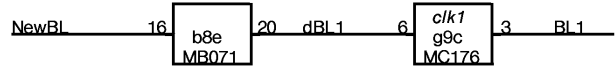
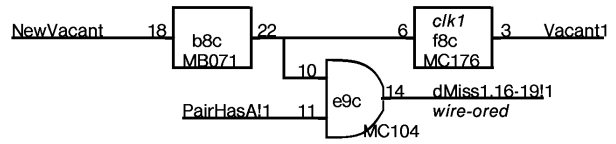
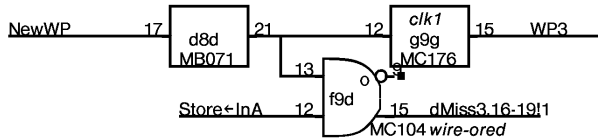
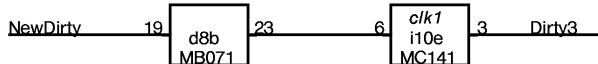
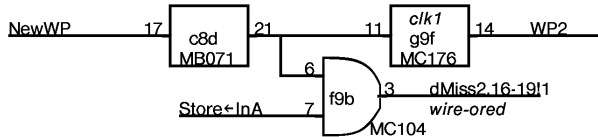
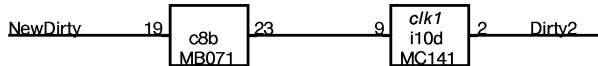
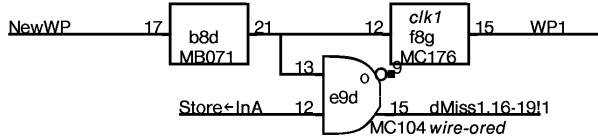
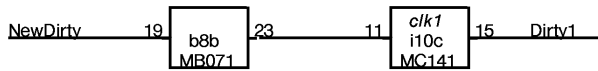
DORADO SCHEMATICS

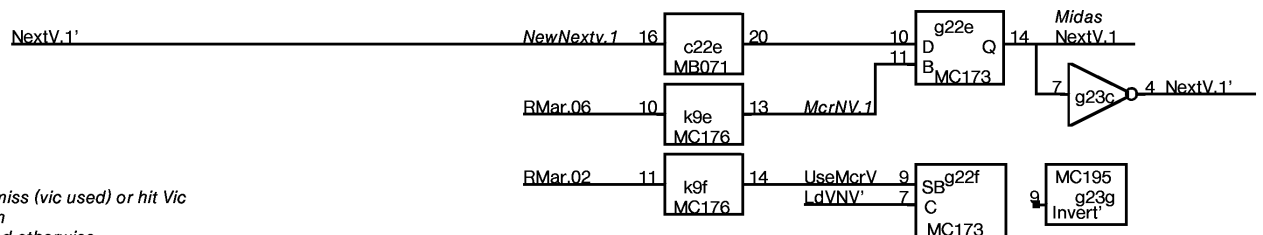
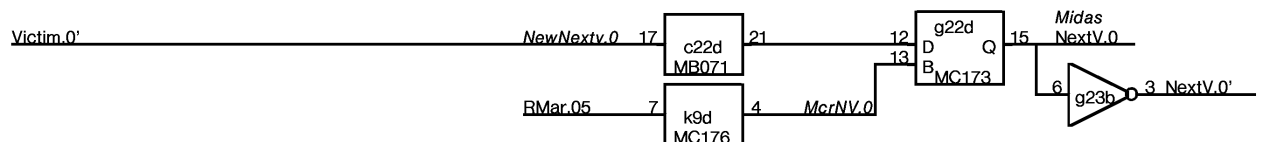
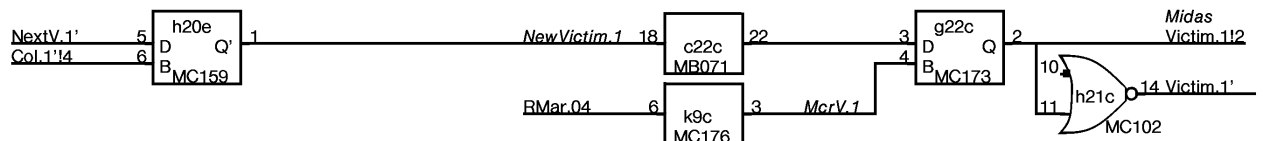
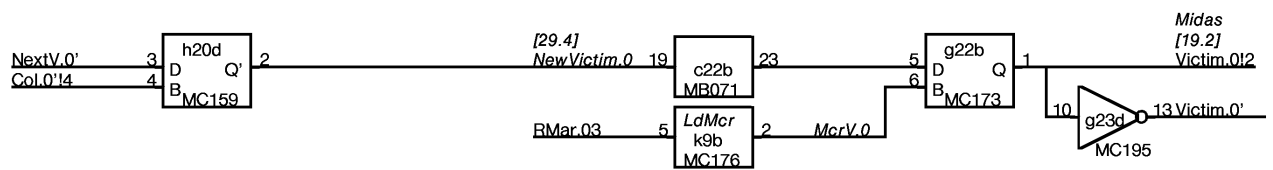
Memory Control

Table of contents

	<u>TITLE</u>	<u>Page</u>
Data paths complete except for repetitions of address bits	Main data path bits 21,27,31; A address bit 7; A memory and comparator bit 6	01
	Cache flags, column 0 and common	02
	Cache flags, columns 1-3	03
	Victim and next victim	04
	Cache A parity, control pipe, Mcr	05
	Mar and BMux drivers and receivers	06
Bit slices for address bits	Main data paths, bits 04-11	07
	12-19	08
	20-25	09
	26-31	10
	Cache A memory and comparators, bits 04-11	11
	12-19	12
	Cache A memory addressing	13
	Pipe and BR addressing; carry logic and data path control	14
Control	Miss and hold	15
	Ref decoding	16
	Pair	17
	Next	18
	FF decoding	19
	Midas control and multiplexors	20
	Clock distribution	21
	Layout	22
	Loading Information	23
	Multiwire rev changes	24

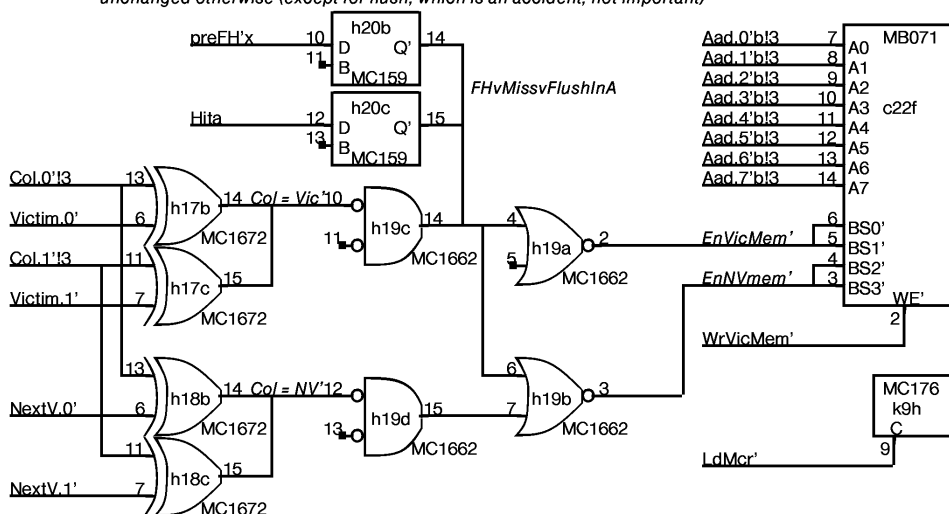






New Vic ← old NV if miss (vic used) or hit Vic
Col if flush
unchanged otherwise

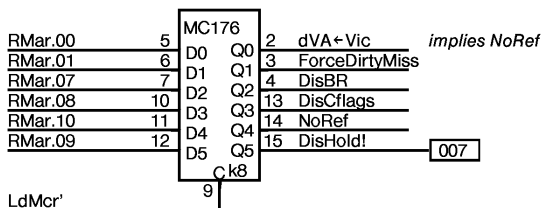
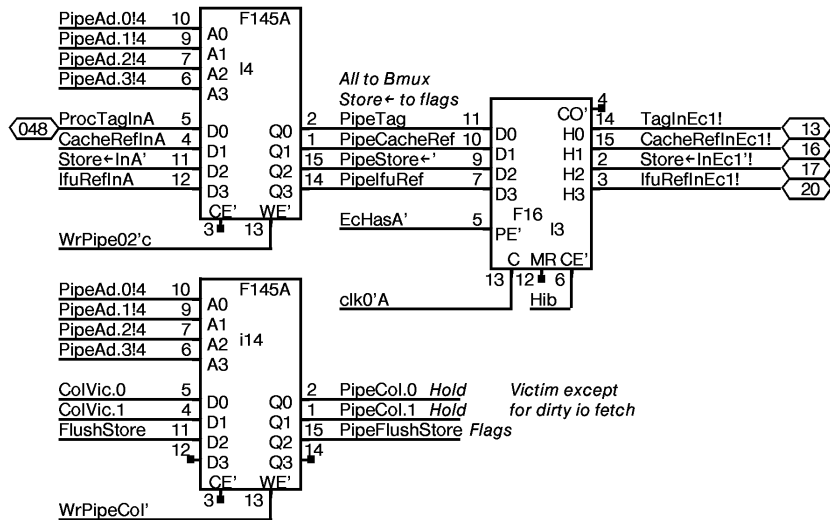
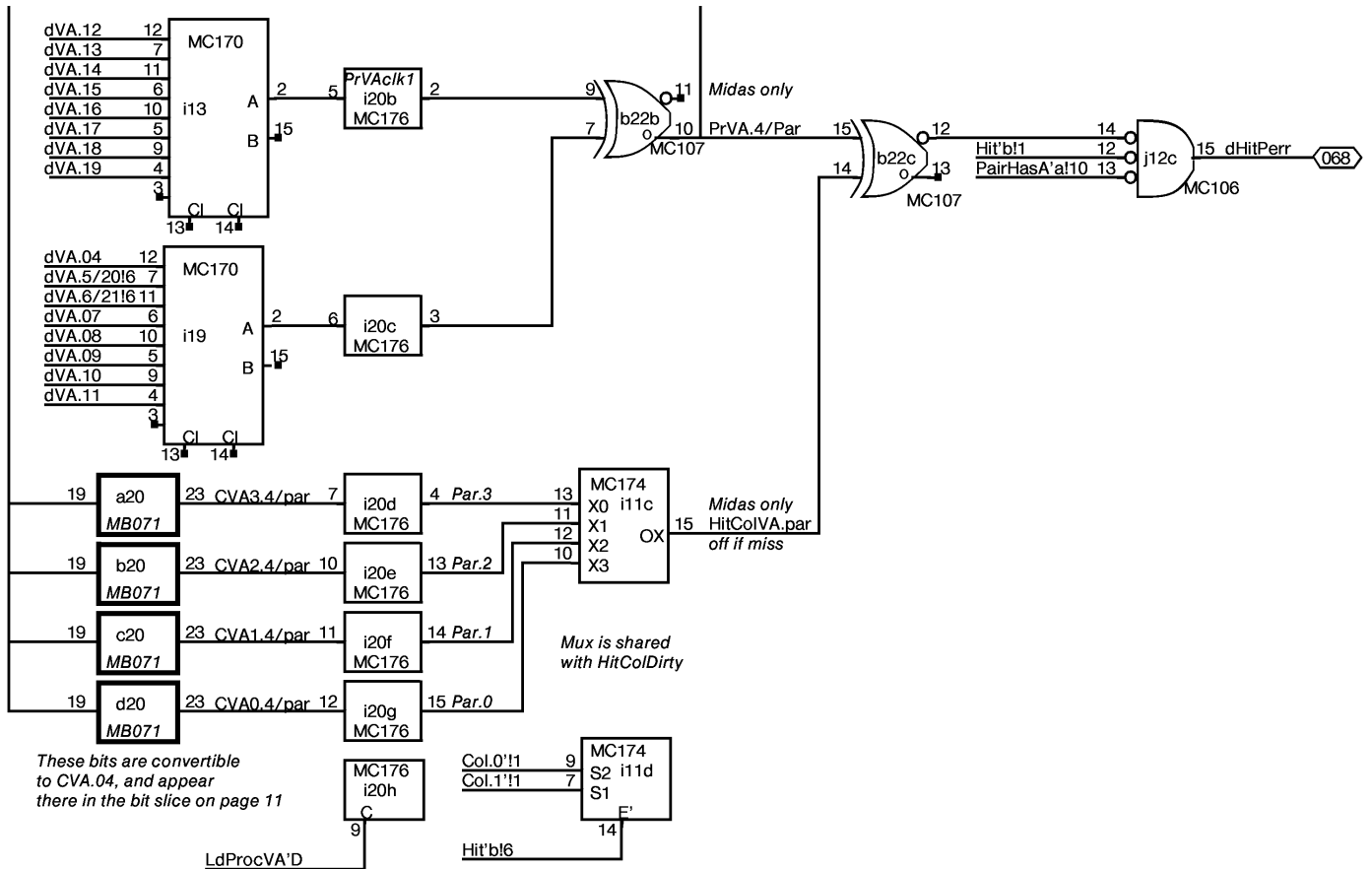
New NV ← something not = old Vic or old NV if miss or hit Vic (NV used) or hit NV
unchanged otherwise (except for flush, which is an accident, not important)

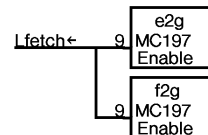
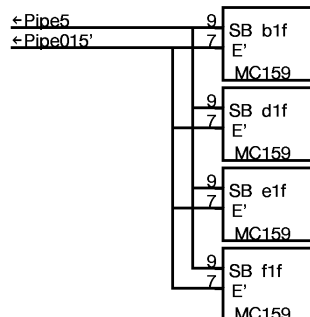
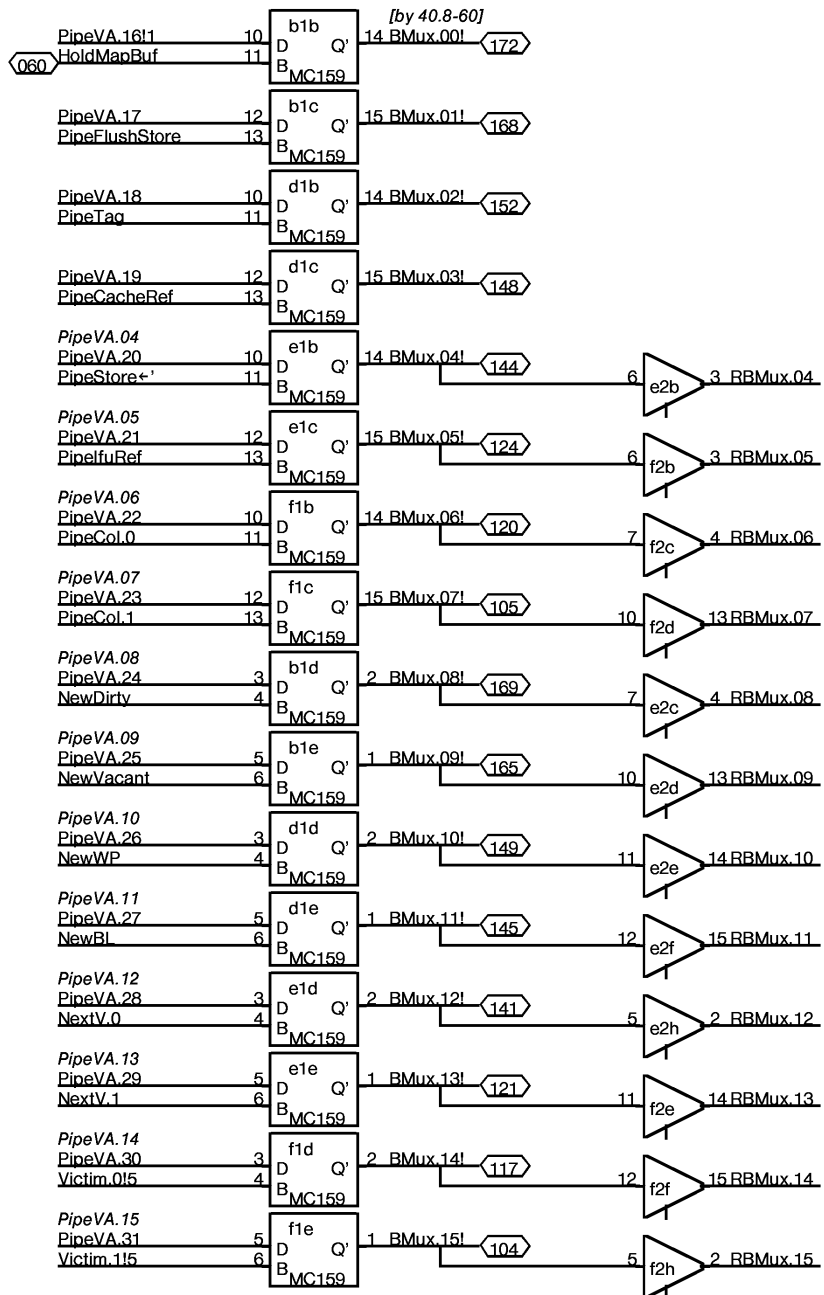
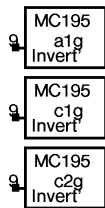
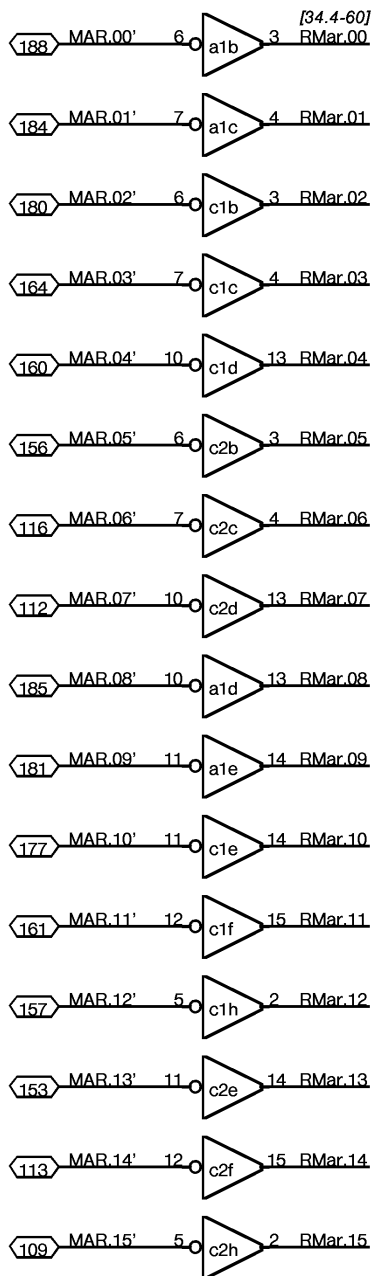


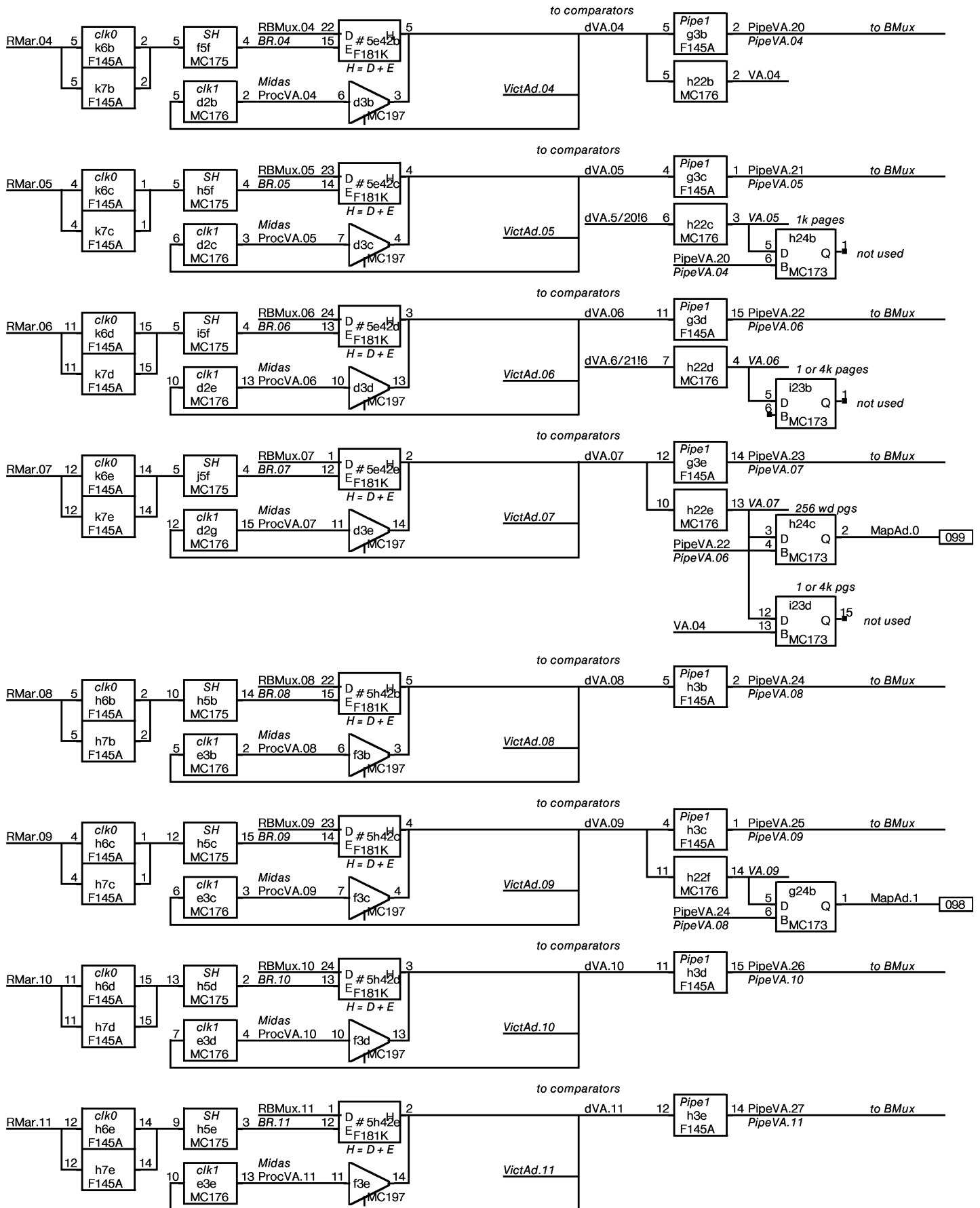
The Victim memory is written on a CacheRef, PreFetch or ltuRef that misses, and on any FlushStore or Flush+.

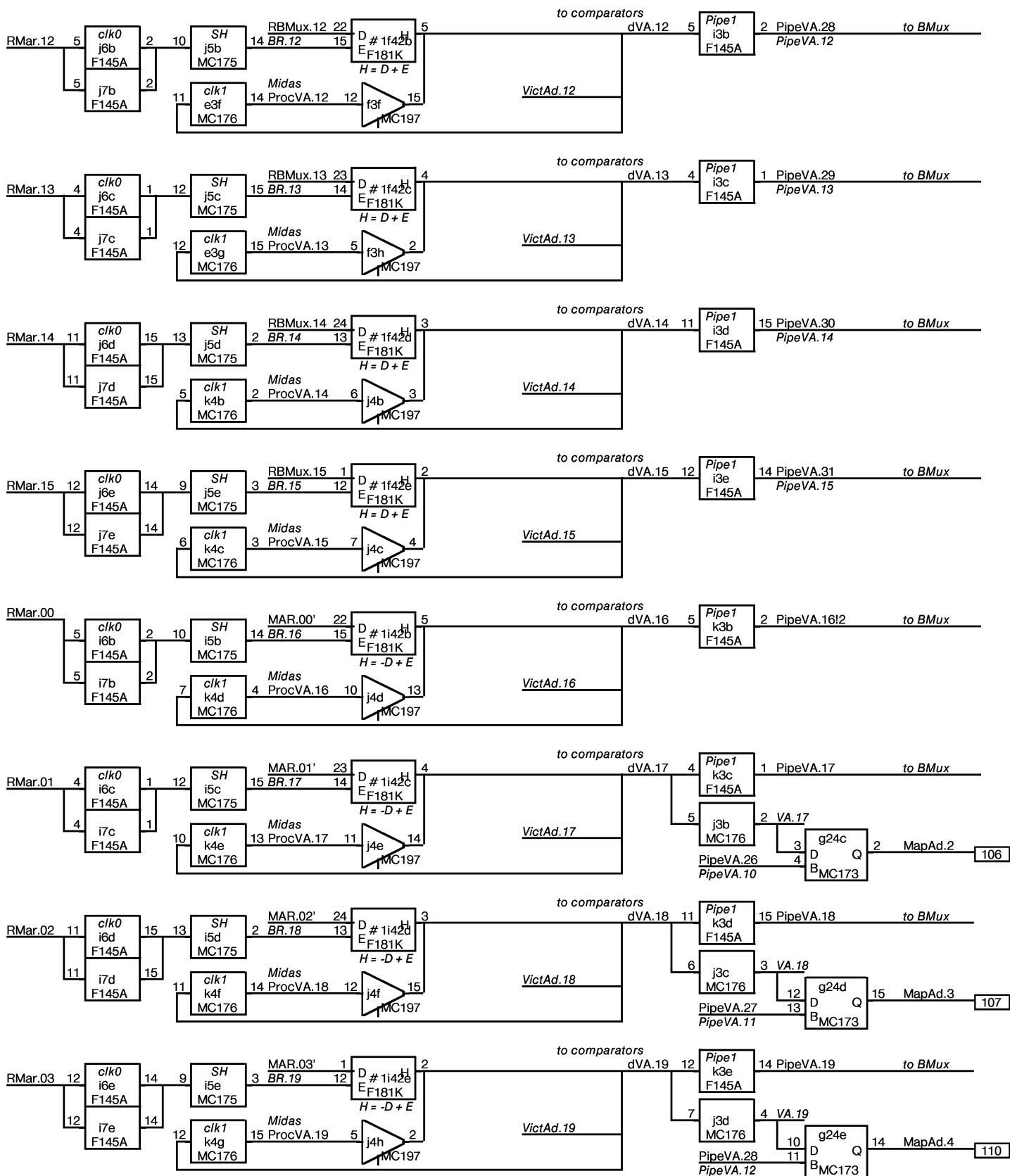
For a Flush+, Victim is first written with 0 on a miss or with the column of the hit. NextV is garbaged at the same time. On a dirty hit, a FlushStore follows, smashing Victim and NextV again.

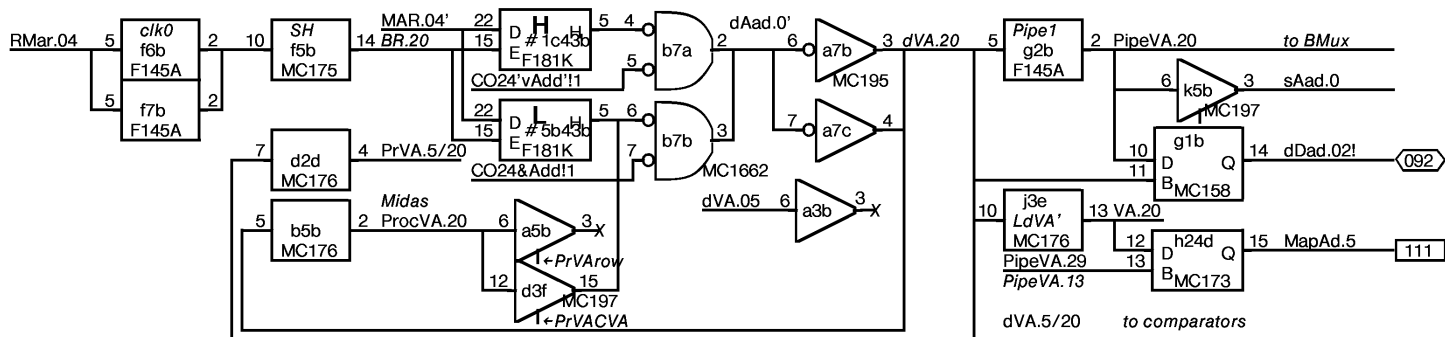
FlushInA
dFlushStore would be better, but the timing is wrong



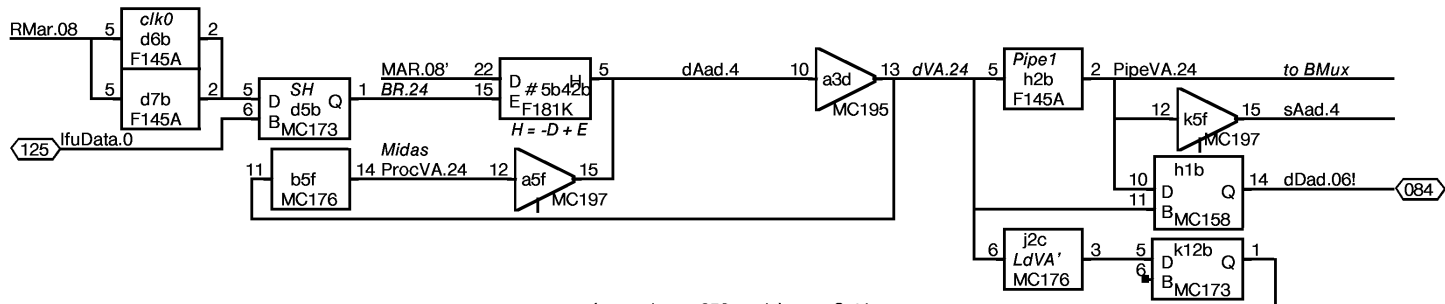
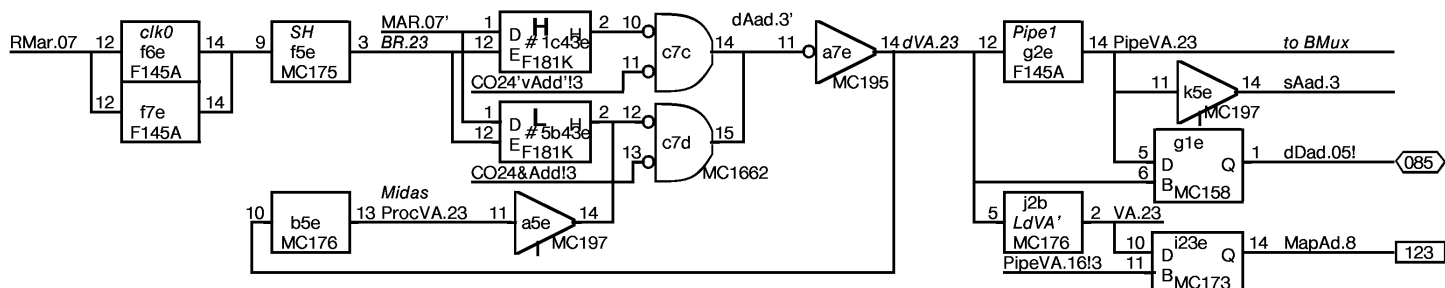
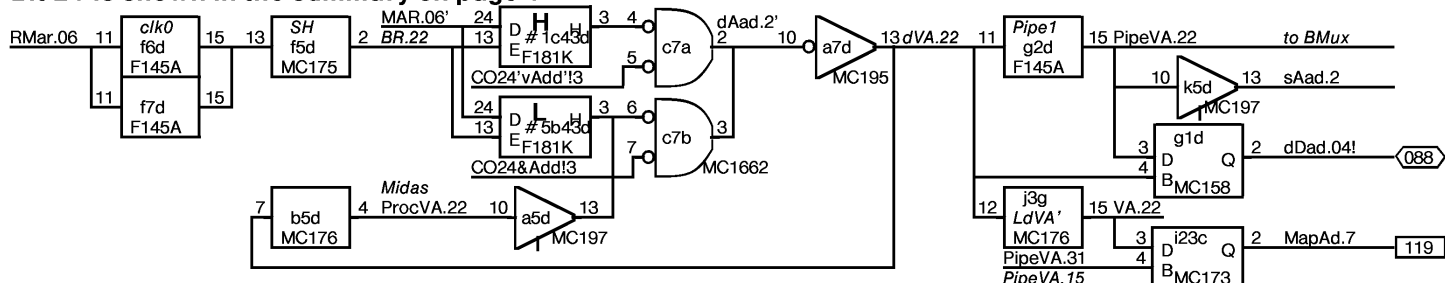






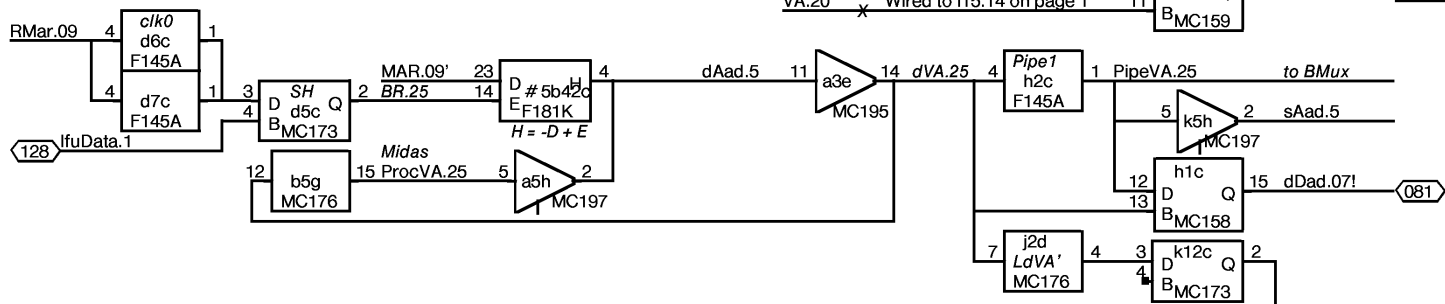


Bit 21 is shown in the summary on page 1



(page size = 256words)

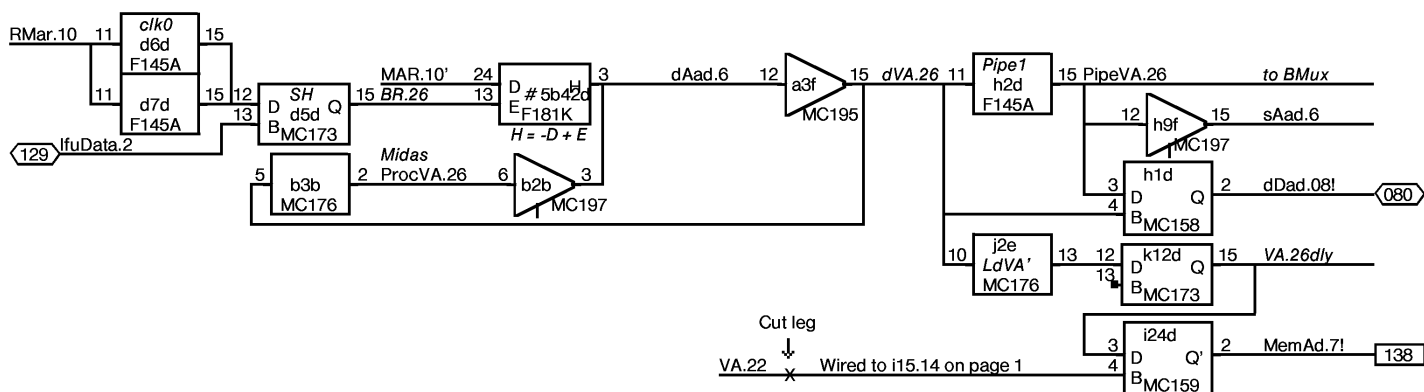
Cut leg



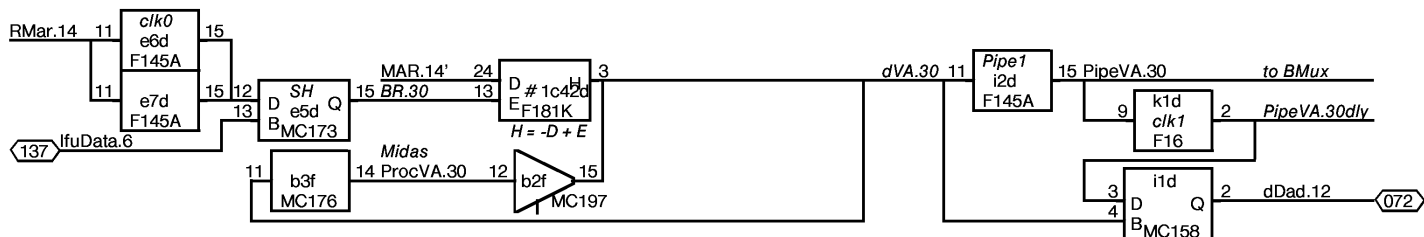
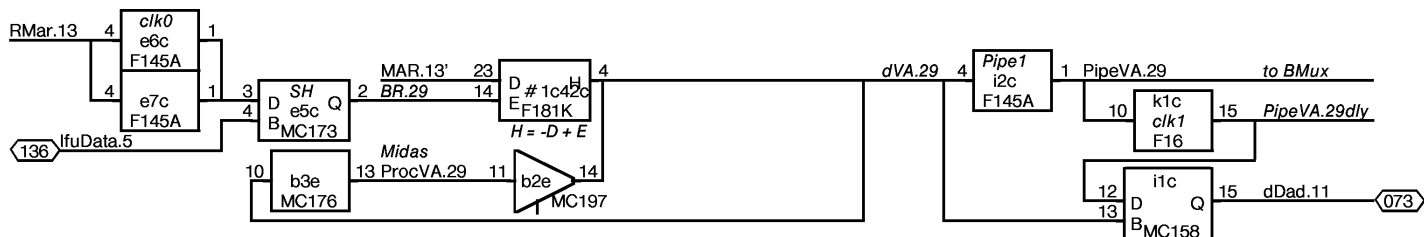
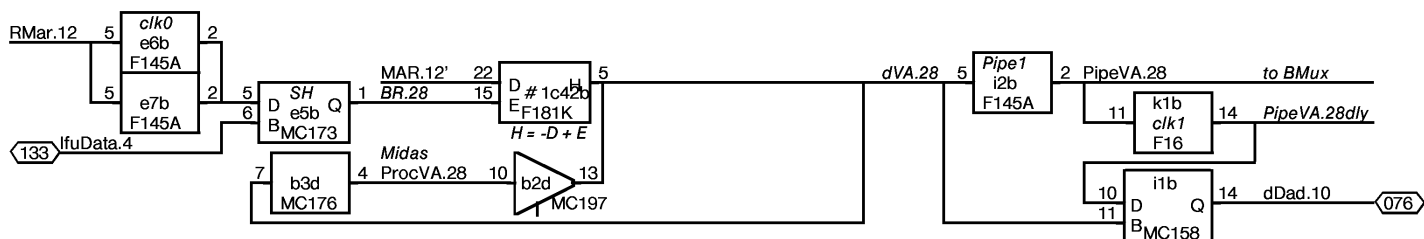
(page size = 256words)

Cut leg

XEROX	Project	Drawing	File	Designer	Rev	Date	Page
PARC	Dorado	Main data paths: 20-26	MemC09.sil	Lampson	Be	7/11/85	09

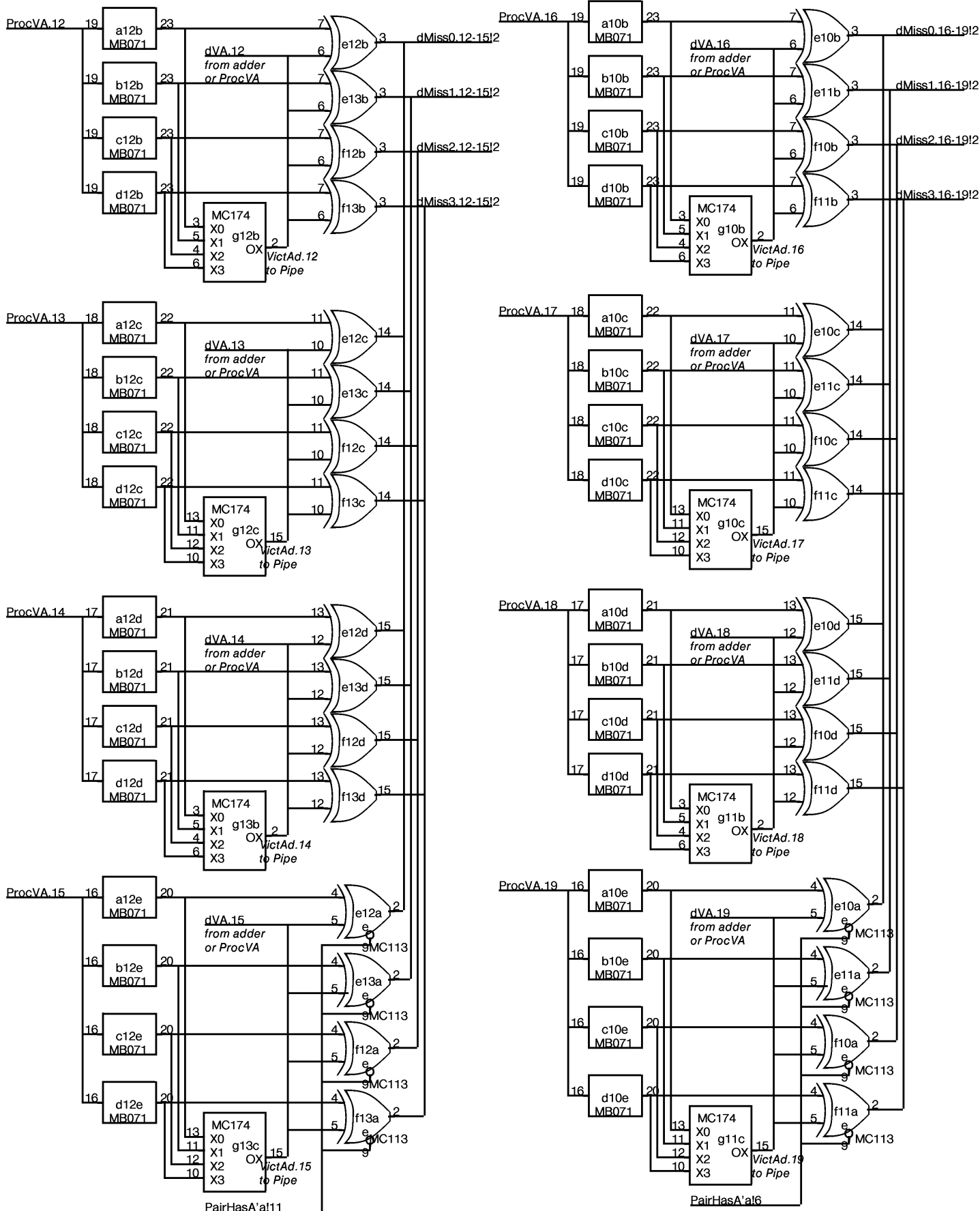


Bit 27 is shown in the summary on page 1



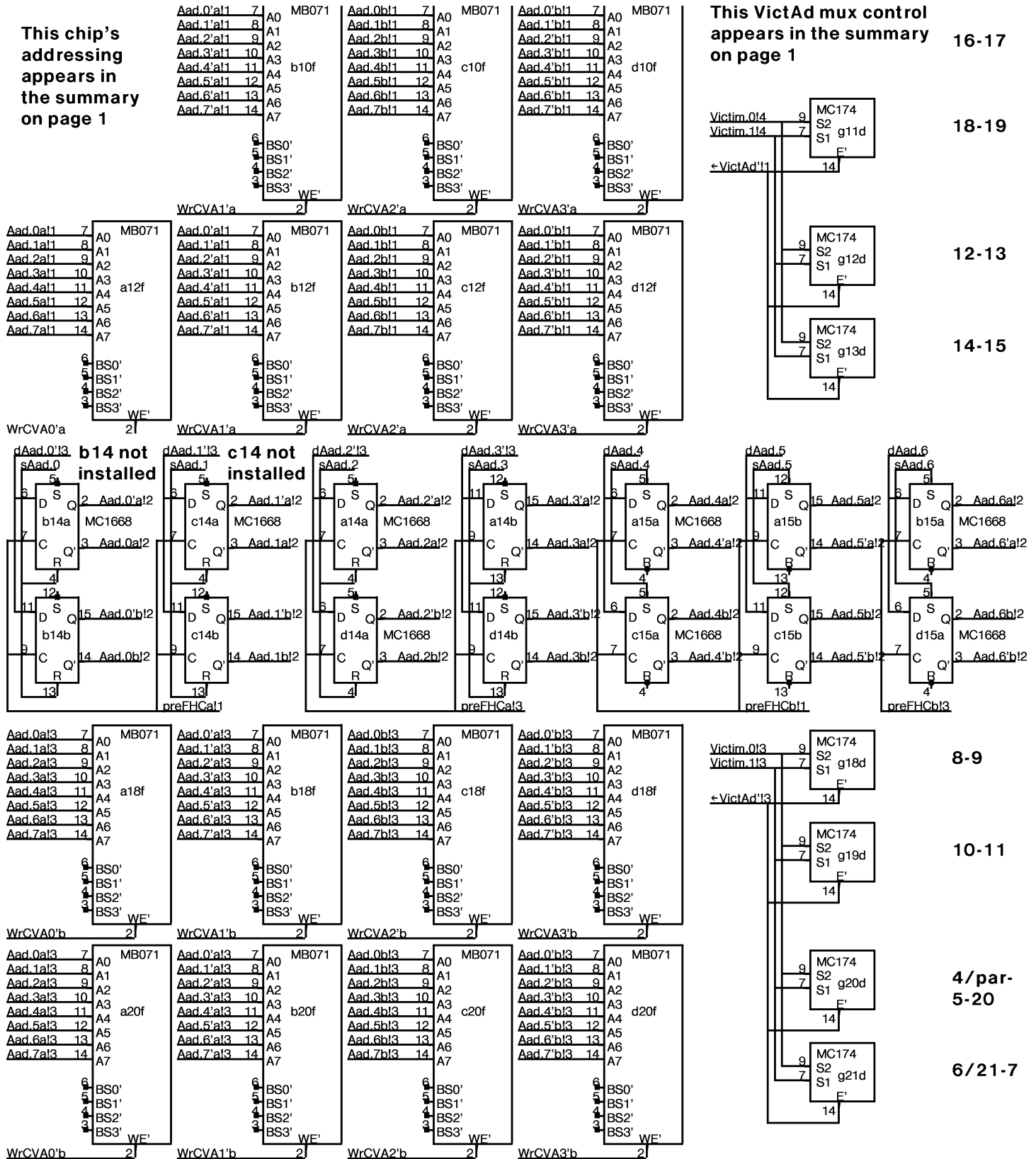
Bit 31 is shown in the summary on page 1

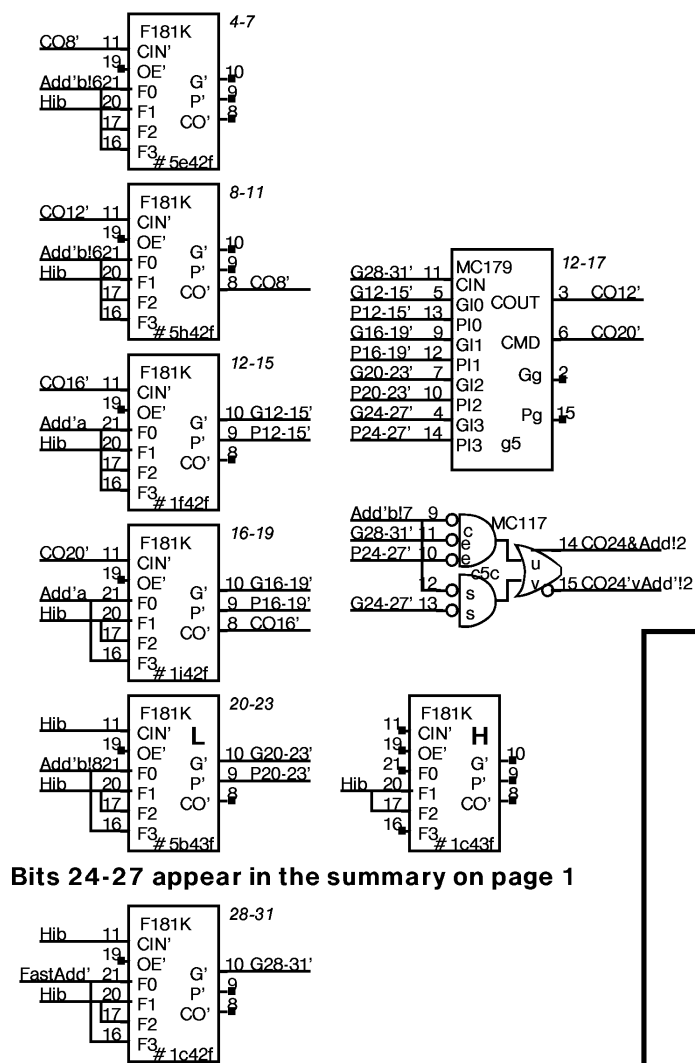
Project	Drawing	File	Designer	Rev	Date	Page
XEROX PARC	Dorado	Main data paths: 26-31	MemC10.sil	Lampson	Be	7/15/85
						10



This chip's
addressing
appears in
the summary
on page 1

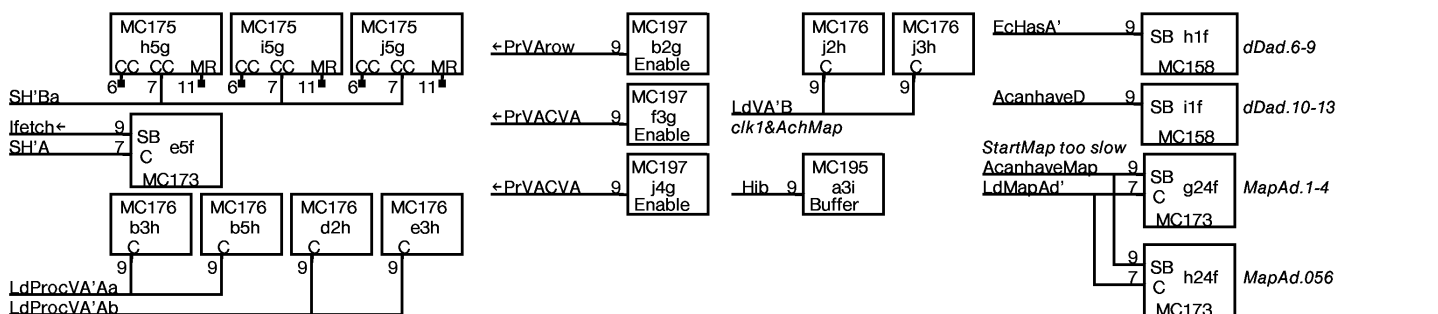
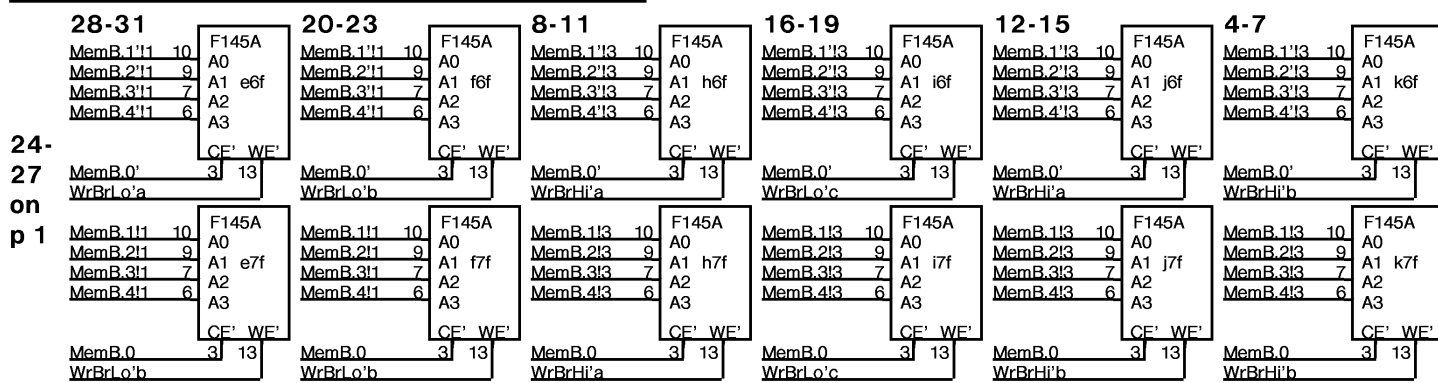
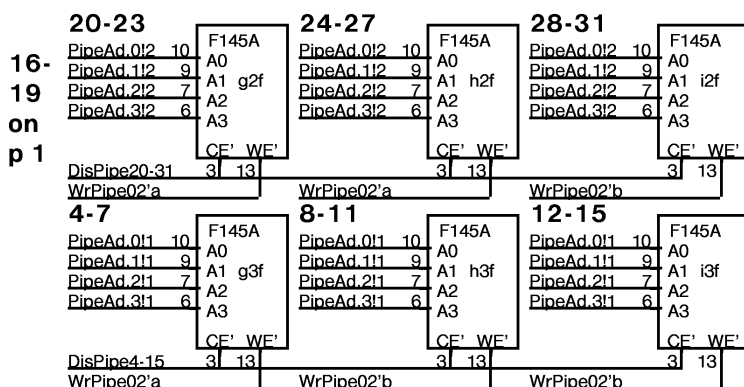
This VictAd mux control
appears in the summary
on page 1

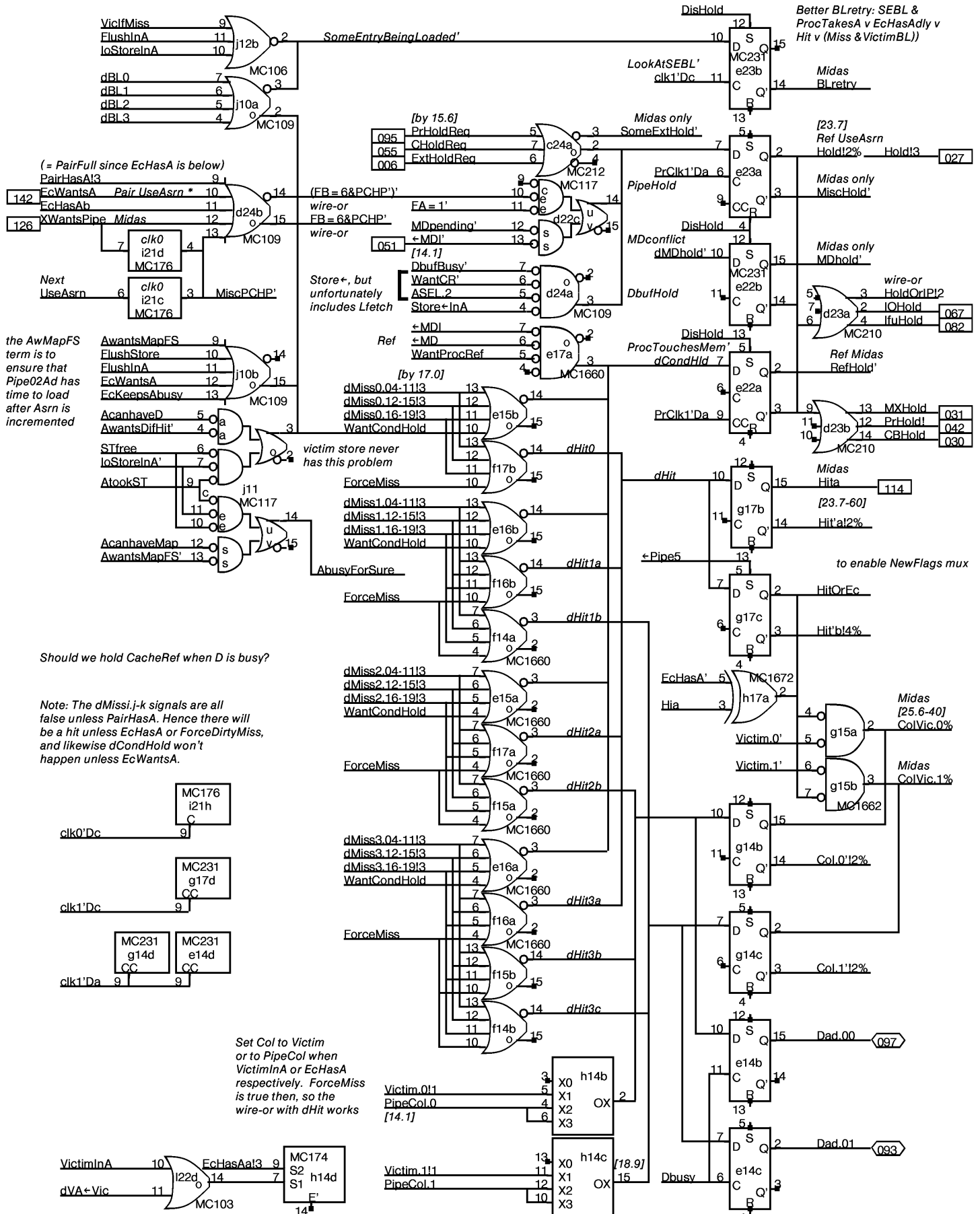




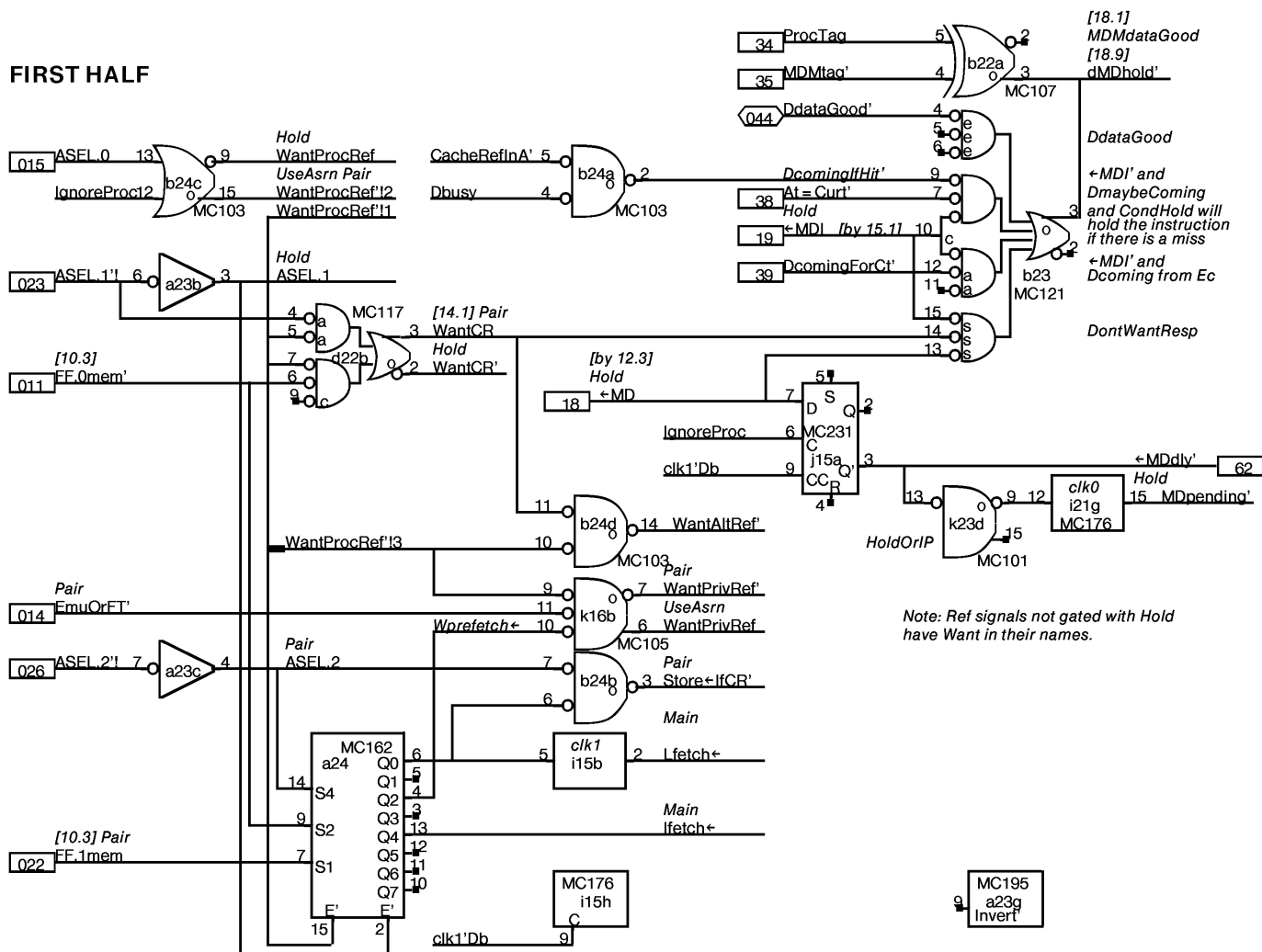
dAad. A RAM outputs are 2.8 + 10 = 12.8 later or at 22.6

dVA, for comparators, must meet the RAM outputs

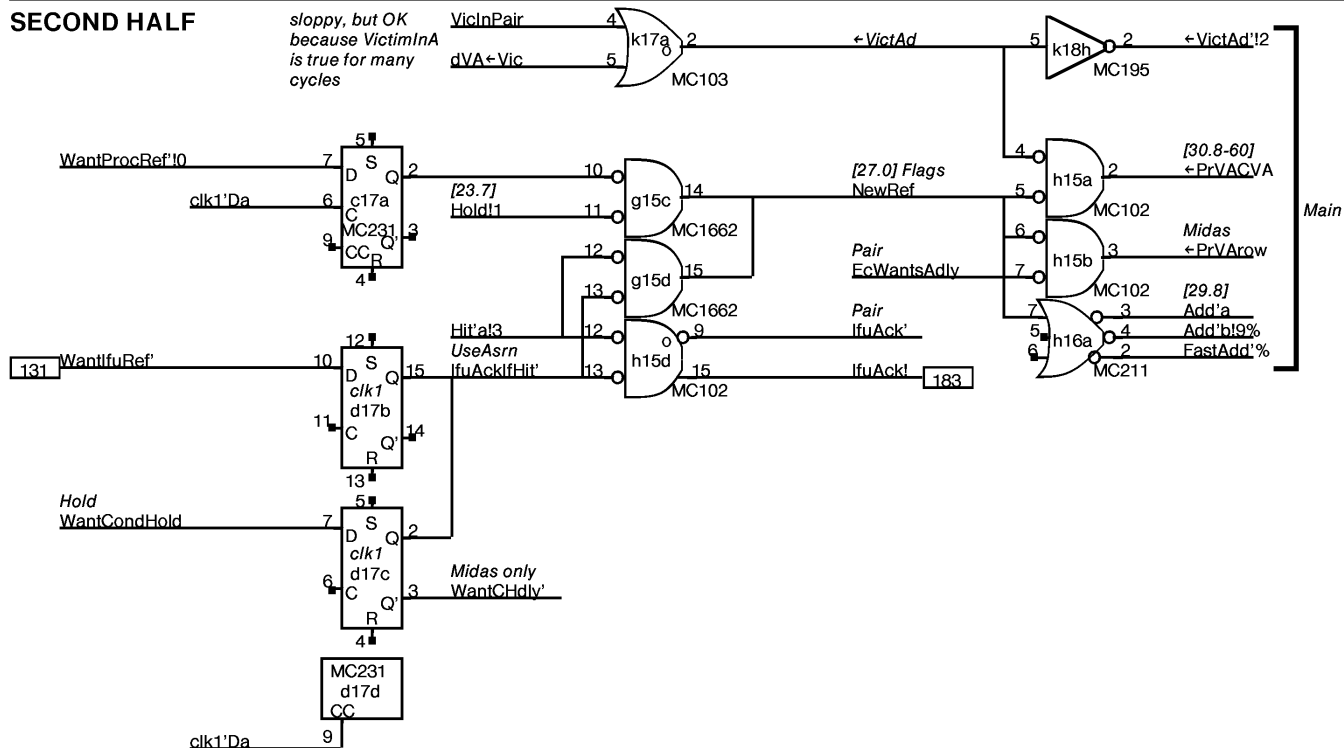


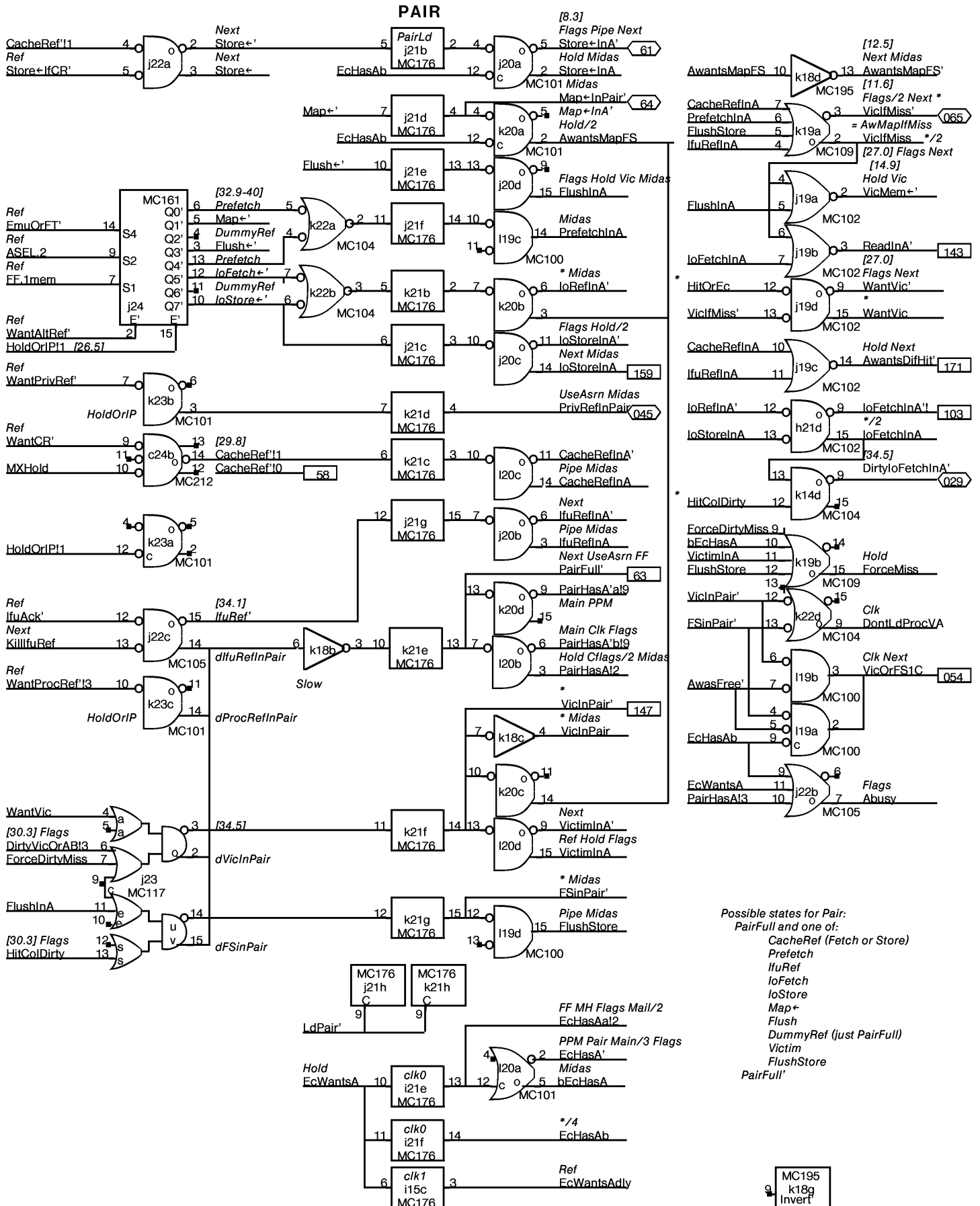


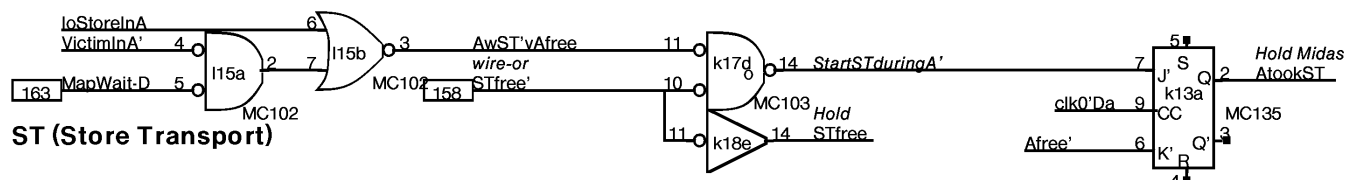
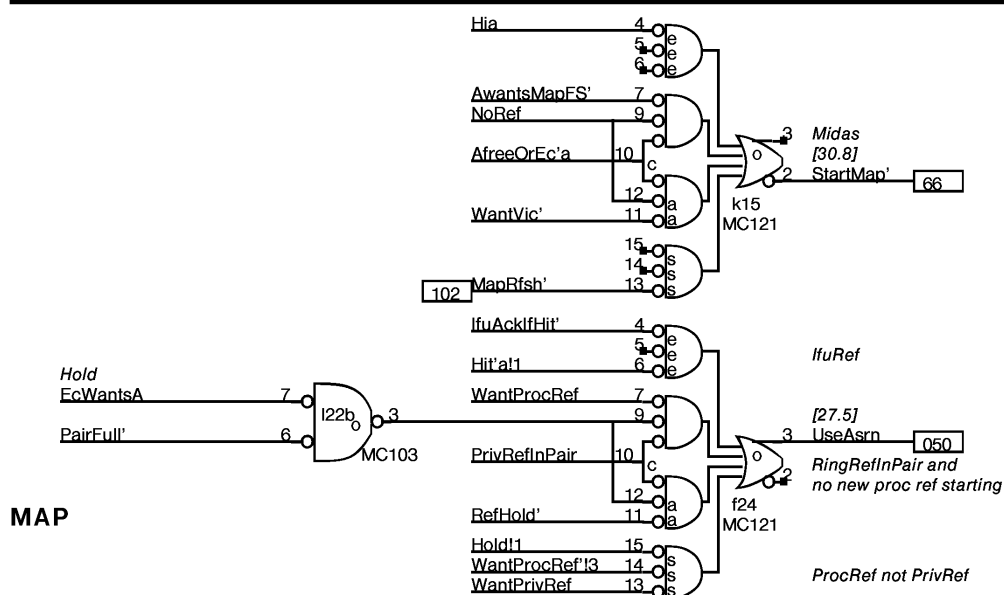
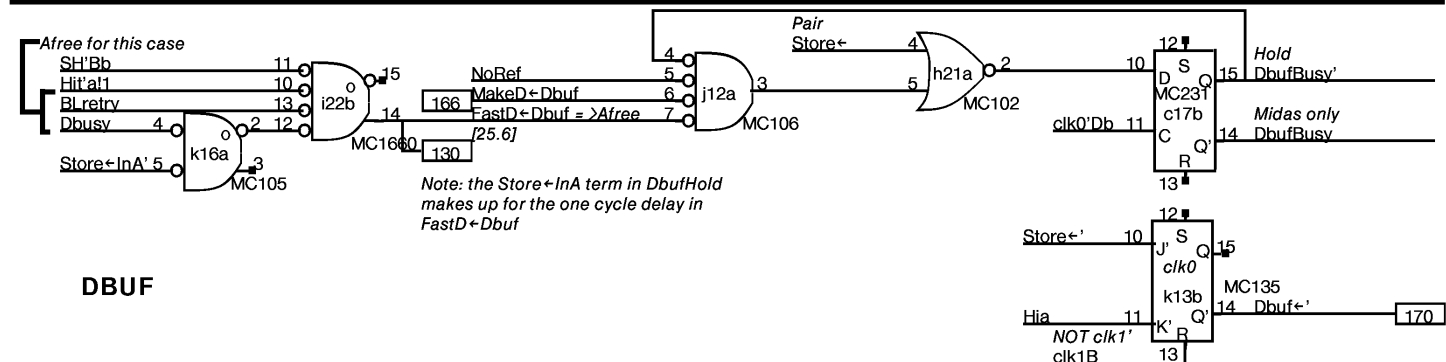
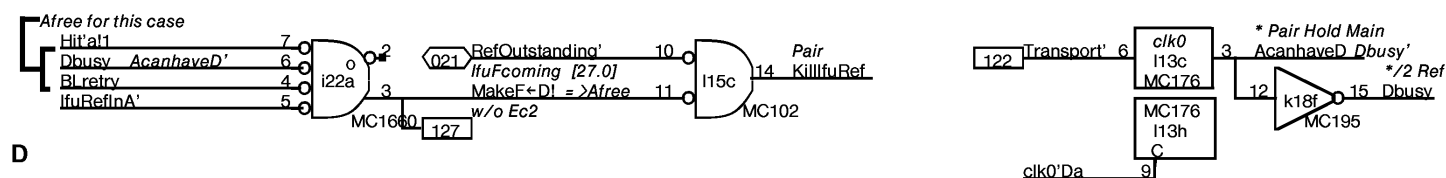
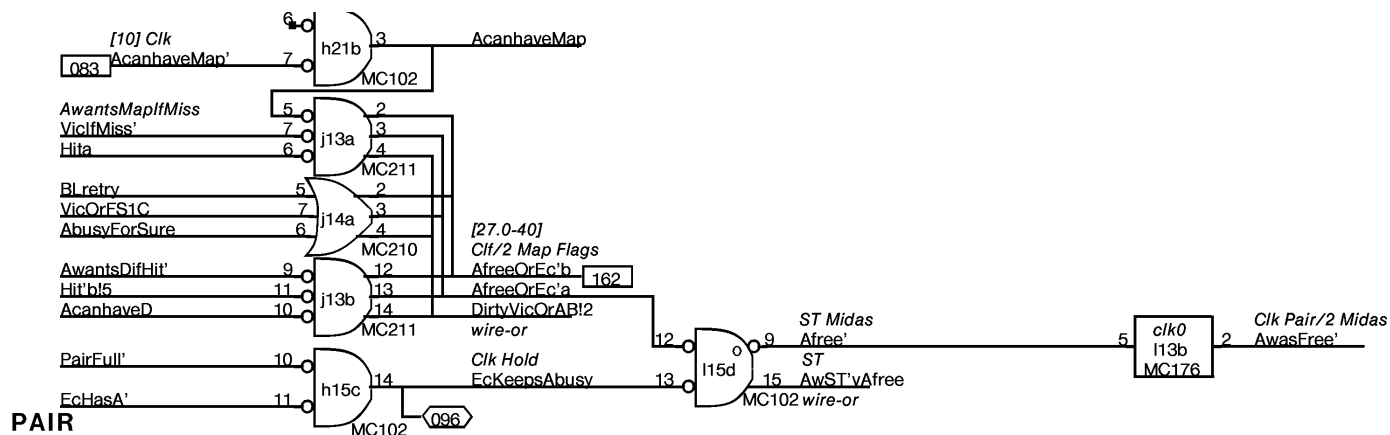
FIRST HALF



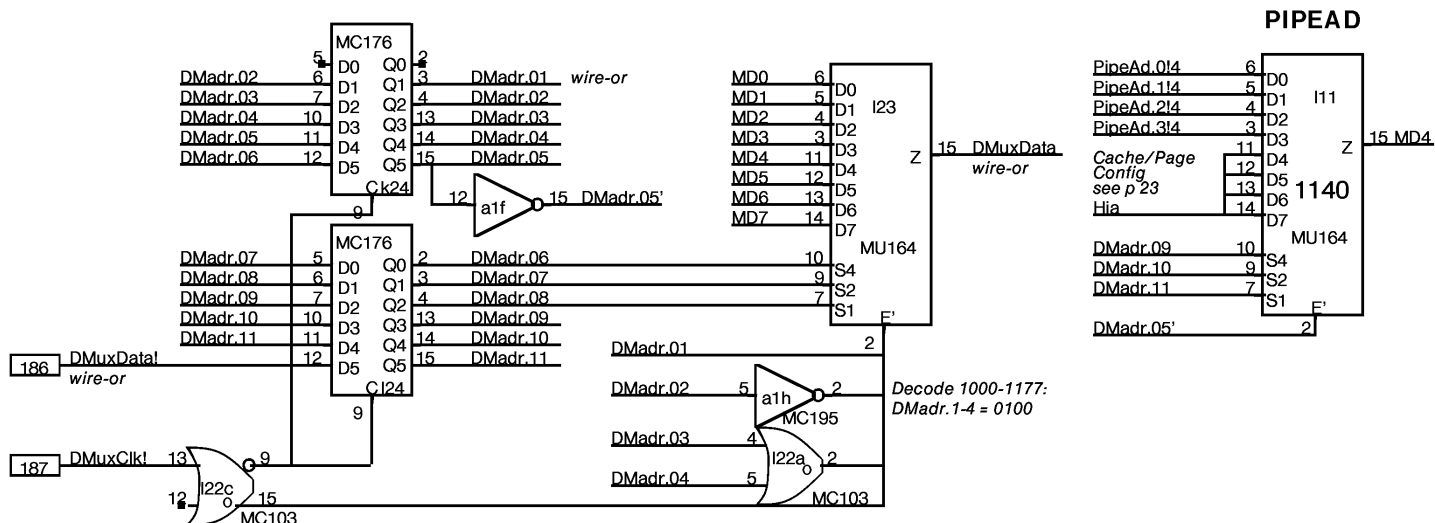
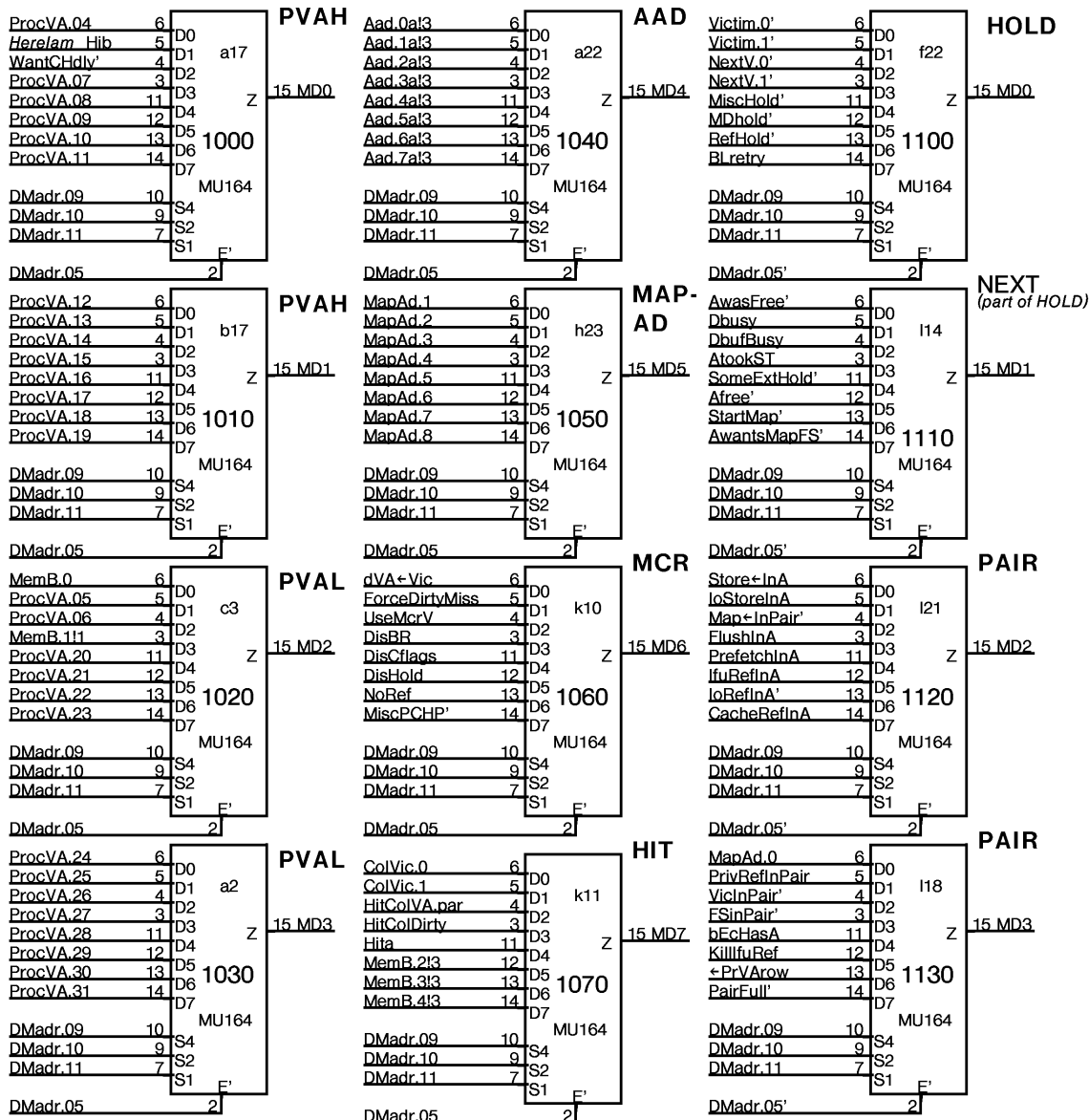
SECOND HALF

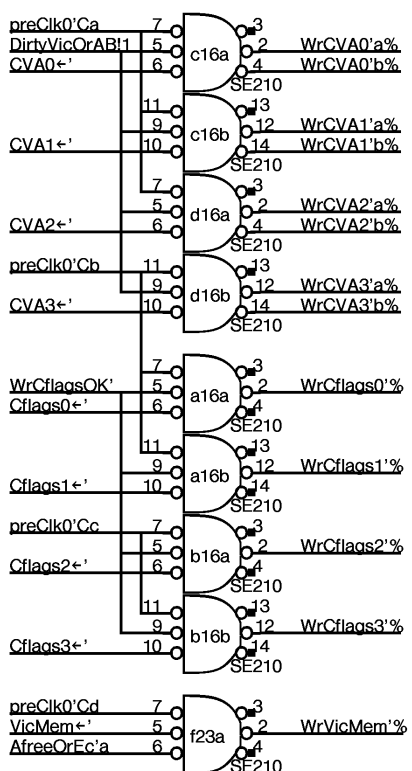
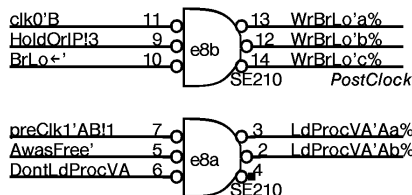
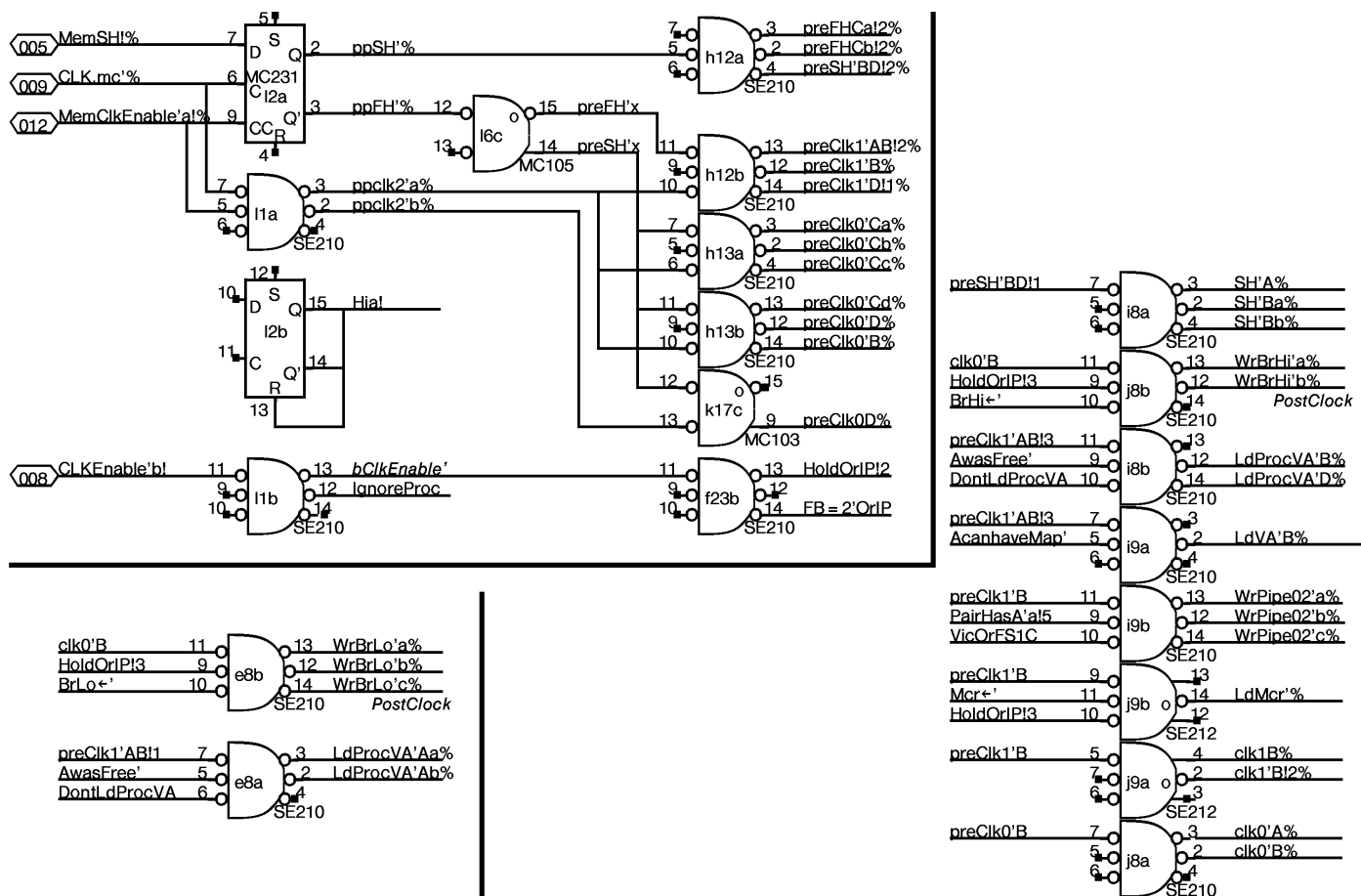




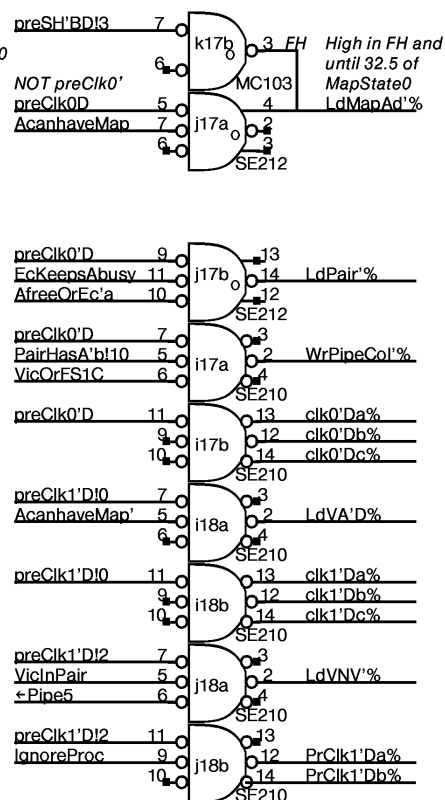


XEROX	Project	Drawing	File	Designer	Rev	Date	Page
PARC	Dorado	Next	MemC18.sil	Lampson	Be	7/17/85	18





MapAd must be
 low by 12.8 before clk0 of StartMap
 to: let row MapAd through by 6 before clk0
 high from clk0 or StartMap to 27.5 after
 to: hold row MapAd
 low by 2.8 before clk0 after StartMap
 to: let col MapAd through by 4 after clk0



Cxxx Mar 0/8 BMux 18 15 IfuData Mar 7/15 BMux 93 dDad.02-10 CLKEn CkSync												CLK.mem 20 RkRsh		
A	a	b	c	d	e	f	g	h	i	j	k	l	B	
1	RMar 0189 DMadr.2,5 195i	BD 0189	RMar 2-4,10-12 195i	BMux driver			2-5 158	6-9 dDad 2-13 158	10-13 158	MemB+37 +Pipe015' 1662	PipeVAdly 28-31 F16	Clock 210	1	
2	MU ProcVA 24-31	Proc 197	5-7,13-15 195i	176	RBMux 4,8,12 5-7,13-15 197			Pipe 20-31 PipeAdl2		VA 141	PipeAdl3	ovh 231	2	
3	dVA buf 22-27 195	VA/dr 26-31 176	MU PrVA56,20-23 MemB.0-1	ProcVA & dr		4-13 5/20,6/21 197		Pipe 4-15 PipeAdl1		17-27 176	Pipe 16-19 PipeAdl4	PipeTag F16	3	
4	24-27 181k	Adder	28-31 181k	4-7 181k	Adder	12-15 181k	8-11 181k	Adder	16-19 181k	ProcVA & dr 197 14 19 176		145	4	
5	ProcVA & dr 197 20 25 176		Cout.24' 117	IfuD/BR 24-31 173		BR latch 4,20-23 175	Cout12,20' 179	5,8-11 175	BR latch 6,16-19	7,12-15 175	sAad.0-5 197	disPipe/3 +Pipe15' 103	5	
6	20-23 181k	Adder	20-23 181k	Base register RAMs			Base register RAMs					MemB.0! Dec, pSHx 105	6	
7	dVAbuf.20-23 dVA.56/2021 195i	dAad.0-3 mux 1662		24-27 145	28-31 145	20-23 145	MemB.1-4 101	8-11 145	16-19	12-15 145	4-7 145	FF PRclk1 176	7	
8	Aada	Aad'a	Aadb	Aad'b	LdProcVA' WrBrHi' 210	Cflags reg 176	Cf+RMar 100	New 174	Clocks		Mcr 176	decod- 162	8	
9					Miss-Cflags 104		Cflags reg BL + 176	sAad 6-7 197				ing PRclk1 176	9	
10	A bits 16-19			Comparators			Victad	Cflags 105	Dirty 141	SEBL WantCH 109	MU Mcr	161	10	
11				16-19			16-19	174	HitColDirty Parity 174	AbusyFS WantCH 117	MU MemB.2-4 +5	MU PipeAd +4	11	
12	A bits 12-15			Comparators			Victad	Pre 210	DirtyVicOrAB 174	dDbufBusy! dHitPerr SEBL 106	VAdly 173	+	12	
13				12-15			12-15	clocks 210	dVA.12-19 parity 170	Afree 211	Dbuf+ AtookST 135	Next/2 MB.0 176.0	13	
14	2/3	0	1	2/3	Dad.0-1 231	Col	Col.0-1 231	PipeCol 174	145	210	DirtyloF!	MU	14	
15	4/5 1668	6/7	4/5	6/7 1668	Hold	1660	ColVic.0-1 NewR,HorEc 1662	IfuAck! +PrVA/2 EckeePSAB 102	Lfatch EcwAdly ABdy MCS 176.1	+MdDly' 231	StartMap 121	Afree! AwST/2 KillfuRef 102	15	
16	210	CVA/Cflags clocks			210	1660	Hit	171	211	102	121	WantPrivRf! FastD+Dbuf 105	+	16
17	MU PrVA 4,7-19 +2		WantPRdly DbufBusy 231	IfuAckIfHit 231	ProcTchMem 1660	1660	Hit	HitOrEc' Col = Vic' 1672	Clocks		+preCik0 MapAdLd StST, +VicA 103	+	17	
18	A bits 8-11			Comparators			Victad	Col = NV' preMCS' New 1672			Pair/3 Next/2 +VictAd 195i	MU Pair	18	
19				8-11			8-11	VNV EnVNV 1662	dVA.4-11 parity 170	WantVic! VicMis AwDirHit 102	ForceMiss VicIfMiss 109	VicOrFS1C (EcHasA) 100	19	
20	A bits 7, 20-21, parity			Comparators			Victad	stuff 159	Parity PrVAclk1 176	101	PairInA (EcHasA)	101	20	
21	or bits 4-7			4,7,20-21 or 4			4,7,20-21 or 4-7	Vic.1'loFiA AchMap dDbufBusy 102	dpCHP,Pr/2 UsaAsrndly MdPend. 176.0	176	Pair	MU Pair	21	
22	MU Aad	dMdH,dPerr dVA.4/par 107	VNV	WantCR,CR dMDc,PipH 117	RefH, MdH 231	MU xHold/3 BLretry VNV	VNV	VA 4-7,9 176	FastD+Dbuf MakeF+D! 1660	St+,IfuRef Abusy 105	Pref,loRef DntLdPrVA! 104	Midas/2! D+Dbuf PcpVic 103	22	
23	ASEL12 mkD+CD 195i	dMDhold 121	RAM	xxHold dMisch 210	Misch, BLrty 231	WrVicMem IP buf 210	NextV' Vic.0' FF-5-7' 195i	MU	78 173	dVic/FS 117	Pair/3,MDd (HoldOrIP) 101	Midas 164	23	
24	WantRef decodes 162	WantPR! DcomingH St+icr,wAR 103	xxHold CacheRef' 212	dDbufHold PCHP 109	FB decode 171	UseAsrn 121	1-4 173	MapAd 056 173	MemAd 159	Ref decodes 161	overhead 176		24	
Cxxx												D		
a	11	b	26	c	39	d	55	e	70	f	86	g	99	
+MDI +MD Asel FF.0-3 PRhold FF.4-7 MemBase MemAd 5-8												h		
XEROX PARC Project Dorado Reference Layout File MemC22.sil Designer Lampson Rev Be Date 7/01/79 Page 22												i		
												j		
												k		
												l		

CACHE CONFIGURATIONS

4k CacheConfig = 3			16k Not used CacheConfig = 2			16k without parity Not used CacheConfig = 1		
Position	Chip	Cut (X) or wire (pin-pin)	Position	Chip	Cut (X) or wire (pin-pin)	Position	Chip	Cut (X) or wire (pin-pin)
a03	195	3x 4x cut 5-6 from 5/20-6/21	a03	195	3-6 4-7 connect 5-6 to 5/20-6/21	a03	195	3-6 4-7 connect 5-6 to 5/20-6/21
a07	195	2-15 3-4 connect 20-21 to 5/20-6/21	a07	195	2x 4x cut 20-21 from 5/20-6/21	a07	195	2x 4x cut 20-21 from 5/20-6/21
a05	197	3x 4x read ProcVA.20-21 for CVA	d03	197	2x 15x read ProcVA.20-21 for row	d03	197	2x 15x read ProcVA.20-21 for row
b14	1668	remove disconnect Aad.0						
c14	1668	remove disconnect Aad.1						
e20	113	3x	e20	113	3x	b22	107	7x 9x 14x 15x 6-10 connect 4 to 4/par disconnect Perr reporting
e21	113	3x	e21	113	3x			
f20	113	3x	f20	113	3x			
f21	113	3x keep parity from comparators	f21	113	3x			
g20	174	2x keep parity from CVA	g20	174	2x			
			l11	164	12x make CacheConfig = 2	l11	164	11x make CacheConfig = 1

PAGE SIZE CONFIGURATIONS

256 words PageConfig = 3			1k words Not used PageConfig = 2			4k words Not used PageConfig = 1		
Position	Chip	Cut (X) or wire (pin-pin)	Position	Chip	Cut (X) or wire (pin-pin)	Position	Chip	Cut (X) or wire (pin-pin)
i24	159	4x 6x 11x 13x 4-6-11-13-9 MemRA+0 0 0 0	i24	159	11x 13x 11-13-9 MemRA+0 0 22 23			MemRA+20 21 22 23
			h24	173	3x 4x 1-2 MapAd.0+4-5 for 6-7	h24	173	10x 12x 10-3 12-5 MapAd.5,6+5,7 for 20,21
			i23	173	3x 10x 3-5 10-12 MapAd.7,8+6,7 for 22,23	i23	173	3x 10x 3-5 10-13 MapAd.7,8+6,4 for 22,23
			l11	164	14x make PageConfig = 2	l11	164	13x make PageConfig = 1

CURRENTLY NONE